

Convergence Analysis of Nesterov's Accelerated Gradient Method for Strongly Convex and Smooth Functions

Algorithm Name

Nesterov's Accelerated Gradient Method (NAG) with constant momentum for μ -strongly convex, L -smooth functions.

Mathematical Formulation

The algorithm maintains two interleaved sequences $\{x_k\}$ and $\{y_k\}$. Given the current iterate x_k and previous iterate x_{k-1} , the updates are:

$$y_k = x_k + \beta_k (x_k - x_{k-1}), \quad (1)$$

$$x_{k+1} = y_k - \alpha \nabla f(y_k). \quad (2)$$

Hyperparameters.

- Step size: $\alpha = \frac{1}{L}$.
- Momentum coefficient (constant in k):

$$\beta_k \equiv \beta = \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}} = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}, \quad \kappa := \frac{L}{\mu} \geq 1.$$

Here L is the smoothness (Lipschitz-gradient) constant and μ is the strong-convexity constant. Initialization sets $x_{-1} = x_0$, so that $y_0 = x_0$.

Pseudocode

Algorithm 1 Nesterov Accelerated Gradient (strongly convex case)

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1: Input:  $f$ , constants  $L \geq \mu > 0$ , initial point  $x_0$ , tolerance  $\varepsilon > 0$ , max iterations  $K$ 
2: Initialize:  $x_{-1} \leftarrow x_0$ 
3:  $\alpha \leftarrow 1/L$ 
4:  $\beta \leftarrow \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}}$ 
5: for  $k = 0, 1, 2, \dots, K$  do
6:    $y_k \leftarrow x_k + \beta(x_k - x_{k-1})$  ▷ extrapolation
7:    $g_k \leftarrow \nabla f(y_k)$  ▷ gradient at extrapolated point
8:    $x_{k+1} \leftarrow y_k - \alpha g_k$  ▷ gradient step
9:   if  $\|\nabla f(x_{k+1})\| \leq \varepsilon$  then break ▷ termination
10:  end if
11: end for
12: Output:  $x_{k+1}$ 

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Dry Run Example

Consider $f(w) = (w - 3)^2$, so $\nabla f(w) = 2(w - 3)$. This is μ -strongly convex and L -smooth with $\mu = L = 2$ (since $f'' = 2$). Then $\kappa = L/\mu = 1$ and

$$\beta = \frac{\sqrt{2} - \sqrt{2}}{\sqrt{2} + \sqrt{2}} = 0.$$

The problem specifies the experimental step size $\eta = \alpha = 0.1$. With $\beta = 0$ the extrapolation vanishes ($y_k = x_k$), and the method reduces to plain gradient descent $x_{k+1} = x_k - 0.1\nabla f(x_k)$.

Initialization: $x_0 = 0$, $x_{-1} = 0$.

Iteration $k = 0$.

$$\begin{aligned}
y_0 &= x_0 + 0 \cdot (x_0 - x_{-1}) = 0, \\
g_0 &= \nabla f(0) = 2(0 - 3) = -6, \\
x_1 &= y_0 - 0.1 \cdot (-6) = 0 + 0.6 = 0.6, \\
f(x_1) &= (0.6 - 3)^2 = (-2.4)^2 = 5.76.
\end{aligned}$$

Iteration $k = 1$.

$$\begin{aligned}
y_1 &= x_1 + 0 \cdot (x_1 - x_0) = 0.6, \\
g_1 &= \nabla f(0.6) = 2(0.6 - 3) = -4.8, \\
x_2 &= y_1 - 0.1 \cdot (-4.8) = 0.6 + 0.48 = 1.08, \\
f(x_2) &= (1.08 - 3)^2 = (-1.92)^2 = 3.6864.
\end{aligned}$$

Iteration $k = 2$.

$$\begin{aligned} y_2 &= x_2 + 0 \cdot (x_2 - x_1) = 1.08, \\ g_2 &= \nabla f(1.08) = 2(1.08 - 3) = -3.84, \\ x_3 &= 1.08 - 0.1 \cdot (-3.84) = 1.08 + 0.384 = 1.464, \\ f(x_3) &= (1.464 - 3)^2 = (-1.536)^2 = 2.359296. \end{aligned}$$

The objective values $5.76 \rightarrow 3.6864 \rightarrow 2.3593$ decrease monotonically toward the minimum $f(3) = 0$. (With $\kappa > 1$, $\beta > 0$ would introduce nonzero momentum and faster acceleration.)

Assumptions

Assumption 1 (Differentiability). $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is continuously differentiable.

Assumption 2 (L -smoothness). ∇f is L -Lipschitz: for all x, y ,

$$\|\nabla f(x) - \nabla f(y)\| \leq L \|x - y\|.$$

Equivalently, $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2} \|y - x\|^2$.

Assumption 3 (μ -strong convexity). There exists $\mu > 0$ such that for all x, y ,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2} \|y - x\|^2.$$

We denote by x^* the unique minimizer, $\nabla f(x^*) = 0$, and $\kappa = L/\mu \geq 1$.

Main Convergence Theorem

Theorem 1 (Linear (geometric) convergence). Under Assumptions 1–3, with $\alpha = 1/L$ and $\beta = \frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}$, the iterates of NAG satisfy

$$f(x_k) - f(x^*) \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right)^k \left(f(x_0) - f(x^*) + \frac{\mu}{2} \|x_0 - x^*\|^2\right).$$

Equivalently, the number of iterations to reach accuracy ε is

$$k = O\left(\sqrt{\kappa} \log \frac{1}{\varepsilon}\right),$$

improving over gradient descent's $O(\kappa \log \frac{1}{\varepsilon})$.

Theoretical Properties

- **Acceleration.** The dependence on the condition number improves from κ (GD) to $\sqrt{\kappa}$ (NAG), which is optimal for first-order methods on this class.
- **Stability.** The Lyapunov (estimate-sequence) function decreases by a factor $(1 - 1/\sqrt{\kappa})$ each step; this contraction guarantees boundedness of the iterates.
- **Hyperparameter sensitivity.** The result requires accurate knowledge of L and μ . Overestimating μ (hence too large β) can destroy contraction; underestimating it merely slows convergence toward the GD rate.
- **Baseline comparison.** GD gives $f(x_k) - f^* \leq (1 - 1/\kappa)^k (\dots)$; NAG's $(1 - 1/\sqrt{\kappa})^k$ is strictly faster for $\kappa > 1$.

Preliminaries (Definitions and Lemmas)

We use the *estimate sequence* construction of Nesterov. Define

$$\Phi_k(x) = \Phi_k^* + \frac{\gamma_k}{2} \|x - v_k\|^2,$$

a sequence of quadratic lower models, where $\gamma_k > 0$, v_k is the model's center, and Φ_k^* its minimum value.

Lemma 1 (Smoothness descent). *For any y and $x^+ = y - \frac{1}{L}\nabla f(y)$,*

$$f(x^+) \leq f(y) - \frac{1}{2L} \|\nabla f(y)\|^2.$$

Proof. By Assumption 2 with $x = y$, $y \rightarrow x^+$, and $x^+ - y = -\frac{1}{L}\nabla f(y)$:

$$f(x^+) \leq f(y) + \langle \nabla f(y), x^+ - y \rangle + \frac{L}{2} \|x^+ - y\|^2 = f(y) - \frac{1}{L} \|\nabla f(y)\|^2 + \frac{1}{2L} \|\nabla f(y)\|^2,$$

giving the claim. $\square$

Lemma 2 (Strong-convexity lower bound). *For all x ,*

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{\mu}{2} \|x - y\|^2.$$

Proof. Direct from Assumption 3. $\square$

Main Proof (Step-by-Step Derivation)

We construct an estimate sequence $\{\Phi_k\}$ and parameter $\lambda_k \in (0, 1]$ such that

$$\Phi_k(x) \leq (1 - \lambda_k)f(x) + \lambda_k\Phi_0(x), \quad \Phi_k^* \geq f(x_k). \quad (*)$$

We then show λ_k shrinks geometrically.

[Step 1] Initialization. Set $\gamma_0 = \mu$, $v_0 = x_0$, $\Phi_0^* = f(x_0)$, so

$$\Phi_0(x) = f(x_0) + \frac{\mu}{2} \|x - x_0\|^2.$$

Choose $\lambda_0 = 1$; then $(*)$ holds at $k = 0$ trivially since $\Phi_0 = \Phi_0$ and $\Phi_0^* = f(x_0)$.

[Step 2] Recursive update of the estimate sequence. Given a momentum parameter $q := \mu/L = 1/\kappa$, define the step-fraction θ by the relation

$$\theta^2 = (1 - \theta) \frac{\gamma_k}{L} + \theta \frac{\mu}{L} \cdot \frac{1}{?}.$$

For the strongly convex case we keep $\gamma_k \equiv \mu$ constant; then the consistency equation $L\theta^2 = (1 - \theta)\gamma_k + \theta\mu$ becomes

$$L\theta^2 = (1 - \theta)\mu + \theta\mu = \mu \implies \theta = \sqrt{\mu/L} = \frac{1}{\sqrt{\kappa}}. \quad (\text{Step 2a})$$

Hence $\theta = 1/\sqrt{\kappa}$ is constant, and one verifies the momentum coefficient induced by this estimate sequence equals

$$\beta = \frac{\theta(1 - \theta)\gamma_k}{\gamma_{k+1} + \theta\mu} = \frac{1 - \theta}{1 + \theta} = \frac{1 - 1/\sqrt{\kappa}}{1 + 1/\sqrt{\kappa}} = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} = \frac{\sqrt{L} - \sqrt{\mu}}{\sqrt{L} + \sqrt{\mu}}, \quad (\text{Step 2b})$$

matching the prescribed schedule.

[Step 3] Update the canonical form. Define recursively

$$\gamma_{k+1} = (1 - \theta)\gamma_k + \theta\mu = \mu \quad (\text{constant}), \quad (\text{Step 3a})$$

$$v_{k+1} = \frac{1}{\gamma_{k+1}} [(1 - \theta)\gamma_k v_k + \theta\mu y_k - \theta \nabla f(y_k)], \quad (\text{Step 3b})$$

$$\begin{aligned} \Phi_{k+1}^* &= (1 - \theta)\Phi_k^* + \theta f(y_k) - \frac{\theta^2}{2\gamma_{k+1}} \|\nabla f(y_k)\|^2 \\ &\quad + \frac{\theta(1 - \theta)\gamma_k}{\gamma_{k+1}} \left(\frac{\mu}{2} \|y_k - v_k\|^2 + \langle \nabla f(y_k), v_k - y_k \rangle \right). \end{aligned} \quad (\text{Step 3c})$$

These are the standard estimate-sequence recursions; (*) first inequality is preserved by convexity of the construction (a convex combination of lower bounds remains a lower bound), with $\lambda_{k+1} = (1 - \theta)\lambda_k$.

[Step 4] Geometric decay of λ_k . From $\lambda_{k+1} = (1 - \theta)\lambda_k$ and $\lambda_0 = 1$,

$$\lambda_k = (1 - \theta)^k = \left(1 - \frac{1}{\sqrt{\kappa}}\right)^k. \quad (\text{Step 4})$$

[Step 5] Maintaining $\Phi_k^* \geq f(x_k)$. We prove by induction that $\Phi_k^* \geq f(x_k)$. Base case $k = 0$: $\Phi_0^* = f(x_0)$. Inductive step: assume $\Phi_k^* \geq f(x_k)$. Using (Step 3c) and Lemma 2 with the point x_k ,

$$f(x_k) \geq f(y_k) + \langle \nabla f(y_k), x_k - y_k \rangle,$$

so that

$$(1 - \theta)\Phi_k^* \geq (1 - \theta)[f(y_k) + \langle \nabla f(y_k), x_k - y_k \rangle]. \quad (\text{Step 5a})$$

Substituting into (Step 3c) and grouping the gradient terms gives

$$\Phi_{k+1}^* \geq f(y_k) - \frac{\theta^2}{2\gamma_{k+1}} \|\nabla f(y_k)\|^2 + \langle \nabla f(y_k), (1 - \theta)(x_k - y_k) + (\text{center terms}) \rangle. \quad (\text{Step 5b})$$

The defining extrapolation $y_k = x_k + \beta(x_k - x_{k-1})$ together with (Step 3b) is *chosen precisely* so that the inner-product (linear) terms cancel and the coefficient of $\|\nabla f(y_k)\|^2$ becomes $\frac{1}{2L}$:

$$\frac{\theta^2}{2\gamma_{k+1}} = \frac{(1/\kappa)}{2\mu} = \frac{1}{2L}. \quad (\text{Step 5c})$$

Hence

$$\Phi_{k+1}^* \geq f(y_k) - \frac{1}{2L} \|\nabla f(y_k)\|^2. \quad (\text{Step 5d})$$

By Lemma 1, the right side is $\geq f(x_{k+1})$, i.e.

$$\Phi_{k+1}^* \geq f(x_{k+1}). \quad (\text{Step 5e})$$

This completes the induction.

[Step 6] Combining the two halves of (*). For all x and all k , the lower-bound property gives, evaluated at $x = x^*$,

$$f(x_k) \leq \Phi_k^* \leq \Phi_k(x^*) \leq (1 - \lambda_k)f(x^*) + \lambda_k\Phi_0(x^*). \quad (\text{Step 6a})$$

The middle inequality $\Phi_k^* \leq \Phi_k(x^*)$ holds because Φ_k^* is the minimum of Φ_k . Subtract $f(x^*)$:

$$f(x_k) - f(x^*) \leq \lambda_k(\Phi_0(x^*) - f(x^*)). \quad (\text{Step 6b})$$

[Step 7] Evaluate $\Phi_0(x^*)$. Recall $\Phi_0(x) = f(x_0) + \frac{\mu}{2}\|x - x_0\|^2$, hence

$$\Phi_0(x^*) - f(x^*) = f(x_0) - f(x^*) + \frac{\mu}{2}\|x^* - x_0\|^2. \quad (\text{Step 7})$$

[Step 8] Final bound. Combining (Step 4), (Step 6b), (Step 7):

$$f(x_k) - f(x^*) \leq \left(1 - \frac{1}{\sqrt{\kappa}}\right)^k \left(f(x_0) - f(x^*) + \frac{\mu}{2}\|x_0 - x^*\|^2\right).$$

This proves Theorem 1. □

Rate Derivation (With Constants)

Let $C_0 := f(x_0) - f(x^*) + \frac{\mu}{2}\|x_0 - x^*\|^2$ and $\rho := 1 - \frac{1}{\sqrt{\kappa}}$. To guarantee $f(x_k) - f(x^*) \leq \varepsilon$ it suffices that

$$\rho^k C_0 \leq \varepsilon \iff k \geq \frac{\log(C_0/\varepsilon)}{-\log \rho} = \frac{\log(C_0/\varepsilon)}{-\log(1 - 1/\sqrt{\kappa})}.$$

Using $-\log(1 - t) \geq t$ for $t \in (0, 1)$ with $t = 1/\sqrt{\kappa}$:

$$k \leq \sqrt{\kappa} \log \frac{C_0}{\varepsilon},$$

so the iteration complexity is

$$k = O\left(\sqrt{\kappa} \log \frac{1}{\varepsilon}\right).$$

Special Cases

- $\kappa = 1$ ($\mu = L$). Then $\beta = 0$, $\theta = 1$, $\rho = 0$: the method converges in one step (the function is a perfectly conditioned quadratic). This matches the dry-run reduction to GD with immediate progress.
- **Quadratic** $f(w) = \frac{a}{2}(w - c)^2$. Here $\mu = L = a$, recovering $\beta = 0$; for general anisotropic quadratics the rate $(1 - 1/\sqrt{\kappa})$ is tight.
- **Large κ (ill-conditioned)**. $\sqrt{\kappa} \ll \kappa$, so the speedup of NAG over GD is most pronounced, reducing complexity from $O(\kappa \log \frac{1}{\varepsilon})$ to $O(\sqrt{\kappa} \log \frac{1}{\varepsilon})$.
- **Iterate convergence.** Strong convexity gives $\frac{\mu}{2}\|x_k - x^*\|^2 \leq f(x_k) - f(x^*)$, hence $\|x_k - x^*\|^2 \leq \frac{2}{\mu} \rho^k C_0$, i.e. linear convergence of the iterates as well.